

Miquel Martí Florences

BATTERY STATE ESTIMATION RESEARCHER · CONTROL AND ENERGY SYSTEMS ENGINEER

Barcelona, Spain

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PhD in battery state estimation and control-oriented modeling, with hands-on experience in experimental energy platforms and hybrid systems.

Summary

Research engineer and PhD focused on control, estimation, and hybrid energy storage systems containing lithium-ion batteries. Experienced in experimental platform development, laboratory support, embedded and powertrain integration, and model-based battery analysis. I combine hands-on engineering with research output, including peer-reviewed publications on state-of-charge estimation and battery modelling.

Experience

CSIC-UPC, Institut de Robotica i Informatica Industrial (IRI)

Barcelona, Spain

PHD CANDIDATE & POSTDOCTORAL RESEARCHER

2023 - Present

- Conduct research on lithium-ion battery state estimation and control-oriented modeling, with a focus on regression-based estimation of internal variables.
- Develop and analyze estimation methods for state of charge and open-circuit voltage under challenging excitation and measurement conditions.
- Contribute to 3 journal publications and 2 submitted manuscripts in battery modeling, estimation, and related energy-system topics.

CSIC-UPC, Institut de Robotica i Informatica Industrial (IRI)

Barcelona, Spain

RESEARCH ENGINEER

2021 - Present

- Maintain and upgrade a hybrid experimental station integrating fuel cells, supercapacitors, and lithium-ion batteries for control and energy research.
- Design control-related hardware, sensor interfaces, and TCP-based communication components to support real-time station operation and data acquisition.
- Contribute to research projects including DOVELAR and Hidrogena Renovable, supporting experimental development and laboratory integration.

UPC Vilanova E3 Team

Vilanova i la Geltrú, Spain

HEAD OF ELECTRONICS AND POWERTRAIN

2016 - 2021

- Student team designing and building electric motorcycles for the MotoStudent competition.
- Designed and integrated the complete electrical system architecture of the electric motorcycle.
- Sized and modeled the battery pack from race energy requirements to support powertrain design decisions.
- Led a 6-7 person electronics and powertrain department, coordinating technical decisions, task execution, and subsystem integration.

Applus+ IDIADA

Santa Oliva, Spain

E-MOTOR TEST ENGINEER INTERN

2019 - 2020

- Supported durability testing activities for electric motor components, contributing to automotive validation work.

AKO Group

Vilanova i la Geltrú, Spain

FIRMWARE TEST ENGINEER INTERN

2019

- Automated firmware test processes in Python for gas detector products, improving testing repeatability and reducing manual effort.

CIMELEC SCCL

Alfarràs, Spain

ELECTRICIAN

2015 - 2020, summers

- Installed and maintained electrical and plumbing systems in residential and industrial projects during summer work periods.

Education

UPC-ETSEIB

Barcelona, Spain

PHD IN AUTOMATION, ROBOTICS AND VISION

2023 - 2026

- Research on regression-based estimation of internal variables in lithium-ion batteries, with emphasis on state-of-charge, model parameter and open-circuit-voltage estimation under challenging excitation conditions.

UPC-ETSEIB

Barcelona, Spain

MASTER'S DEGREE IN INDUSTRIAL ENGINEERING

2020 - 2023

- Master's thesis: control architecture for a microgrid composed of a fuel cell, a battery, and a supercapacitor. Grade: 10/10.

UPC-EPSEVG

Vilanova i la Geltrú, Spain

BACHELOR'S DEGREE IN INDUSTRIAL ELECTRONICS AND AUTOMATION ENGINEERING

2016 - 2020

- Bachelor's thesis: control analysis and EV application of a hybrid storage system based on lithium batteries and supercapacitors. Grade: 9/10.

Skills

Battery & Control	Lithium-ion battery modeling, state estimation, observers, control systems, hybrid energy systems
Programming	Python, MATLAB/Simulink, C (small applications), LaTeX
Lab & Hardware	LabVIEW, cRIO (real-time and FPGA), sensor integration, Altium (small PCBs), oscilloscope, multimeter, soldering
Languages	Catalan (native, C1), Spanish (native), English (C1), French (B1), German (A2)

Publications

Finite-Time Adaptive Observation of State-of-Charge and Open-Circuit Voltage Curves in Lithium-Ion Batteries

SUBMITTED TO AUTOMATICA, 2026

- Marti-Flores, M.; Cecilia, A.; Romero, J.G.; Ortega, R.; Bobtsov, A.; Costa-Castelló, R.

Nonlinear Asymmetric Coding Scheme for Covert Attack Detection in DC Microgrids

SUBMITTED TO IEEE TRANSACTIONS ON SMART GRID, 2026

- Chen, T.; Cecilia, A.; Marti-Flores, M.; Wang, L.; Costa-Castelló, R.; Liu, Z.

OCV Estimation in Electrochemical Batteries with an Unknown Linear Model and Finite Excitation

JOURNAL OF ENERGY STORAGE, 2025

- Marti-Flores, M.; Cecilia, A.; Ortega, R.; Bobtsov, A.; Costa-Castelló, R. DOI: 10.1016/j.est.2025.117665

SoC Estimation in Lithium-Ion Batteries with Noisy Measurements and Absence of Excitation

BATTERIES, 2023

- Marti-Flores, M.; Cecilia, A.; Clemente, A.; Costa-Castelló, R. DOI: 10.3390/batteries9120578

Modeling and Estimation in Lithium-Ion Batteries: A Literature Review

ENERGIES, 2023

- Marti-Flores, M.; Cecilia, A.; Costa-Castelló, R. DOI: 10.3390/en16196846